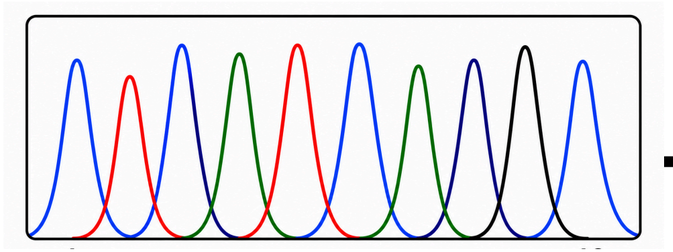
Structural Variants Pipeline

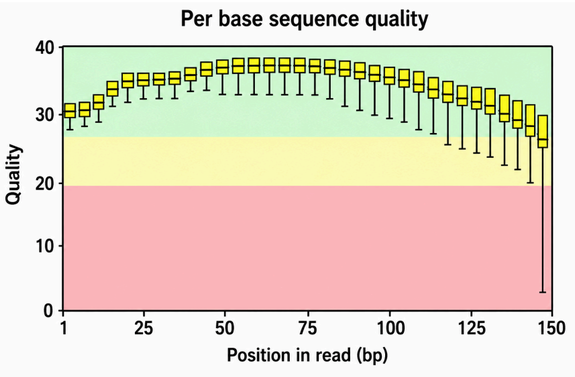


PEARL MILLET



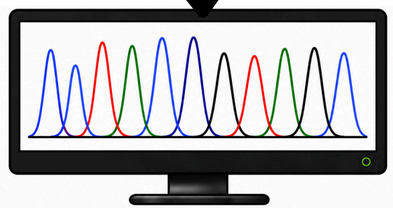
Nucleotide DNA sequencing

Fastp



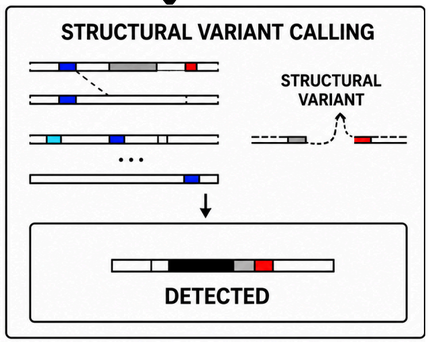
Quality check and adapter trimming

BWA-MEM2



Sequence alignment with reference genome

GATK



SnpEff

STRUCTURAL VARIANT ANNOTATION

Functional Annotation

Functional Annotation Database

Variant ID: SV_LIP_000123
Chromosome: 3
Position: 1,234,567–1,250,890
Type: Deletion
Length: 16,324 bp

Gene Annotation

Gene ID	Gene Name	Transcript	Impact
LOC_04UP1	TAG lipase	LIP1.1	HIGH

Functional Description

Triacylglycerol lipase (EC 3.1.1.3)

Hydrolyses triacylglycerols into glycerol and free fatty acids involved in lipid mobilization and rancidity-related lipid breakdown

Rancidity focused filtering



Rancidity associated genes